SAM-Driven Consistency Regulation for Semi-Supervised Remote Sensing Change Detection

Liang Lv, *Student Member, IEEE*, Zilin Qin, Xuefei Li, and Lefei Zhang, *Senior Member, IEEE*

***Abstract*—Remote sensing image change detection (CD) aims at identifying differences between bi-temporal or multitemporal images. To reduce the reliance of CD networks on large-scale annotated data, semisupervised learning has attracted increasing interest. Most existing consistency learning approaches are based solely on the intrinsic knowledge of the model to refine pseudolabel quality. However, this self-reliant paradigm inherently restricts the cognitive boundaries, resulting in only limited improvements in pseudolabel reliability. Under limited supervision, it often produces unstable predictions. To solve this, we propose a segment anything model-driven consistency regulation (SAM-CR) framework for semisupervised CD (SSCD), which converts model outputs into hybrid cues for specific changed instances and introduces SAM in consistency regulation to reshape the high confidence regions of pseudolabels. Using SAM's rich prior knowledge, the framework dynamically filters out spurious change signals in the unsupervised branch and effectively corrects prediction biases between models. Additionally, to fully exploit the enhanced pseudolabels, we design a multiscale channel similarity (MSCS) strategy within the instance feature alignment (IA) module, enforcing alignment of perturbed instance features and yielding a more reliable supervisory signal. Extensive experiments on three remote sensing CD datasets, including LEVIR-CD, WHU-CD, and CDD-CD, show that SAM-CR outperforms state-of-the-art SSCD methods by approximately 1.5% IoU. The codes are available at https://github.com/QinZelin/SAM-CR**

***Index Terms*—Consistency regulation, remote sensing, segment anything model, semisupervised change detection (CD).**

I. Introduction

With the generation and analysis of massive remote sensing data [1], deep learning (DL) has been widely adopted in various remote sensing domains [2], including semantic segmentation [3], [4] object detection [5], [6], and change detection (CD) [7], [8], among others. CD aims to identify pixel-level differences between bitemporal or multitemporal images and has been applied in many remote sensing tasks such as earth observation, urban development monitoring, and disaster assessment. Moreover, by integrating remote sensing data with domain-specific knowledge, CD has also become an indispensable analytical tool for a wide range of interdisciplinary applications.

Driven by the rapid advancement of DL technologies in remote sensing, CD methods [9], [10], [11] have achieved substantial progress in recent years. Representative architectures such as convolutional neural networks (CNNs) [12] and vision transformers (ViTs) [13] have become dominant paradigms in CD research. Nevertheless, a considerable portion of remote sensing data in practical scenarios remains unlabeled, and manual annotation for supervised learning is both time-consuming and resource-intensive. This limitation poses significant challenges for practical deployment and constrains the further development of CD models. Consequently, semisupervised CD (SSCD) methods [14], [15], [16], [17] have attracted increasing attention, aiming to exploit abundant unlabeled data to enhance model robustness and generalization.

In recent years, increasing attention in SSCD research has been directed toward consistency learning-based approaches. These methods promote the learning of invariant feature representations by introducing various perturbations to the same input image and constraining the model to produce consistent predictions across them. A representative baseline involves transferring the FixMatch algorithm [18], originally proposed for semantic segmentation, to the CD task. This method generates pseudolabels from weakly augmented unlabeled images and applies a confidence-based selection strategy to preserve reliable regions for supervision. On this basis, FPA [19] incorporates rotation-consistency learning in the unsupervised branch to mitigate rotational discrepancies caused by varying target orientations in remote sensing imagery. AdaSemiCD [20] refines pseudolabel quality by adaptively replacing highly uncertain regions in unlabeled data with reliable samples, thereby improving the robustness and accuracy of the learning process. In addition, LGTC [21] employs a three-level consistency learning strategy at the pixel, region, and image levels, which delivers complementary supervision from fine-grained to global semantic scales, strengthening the model's representation learning capability for unlabeled data.

As illustrated in Fig. 1(a), the pseudolabels generated in unsupervised branches often exhibit significant discrepancies from the ground-truth labels. Enhancing the reliability of these pseudolabels remains one of the primary challenges in SSCD. Most existing approaches rely solely on the model's intrinsic knowledge to improve pseudolabel quality. However, this self-reliant strategy inherently restricts the optimization scope, leading to limited gains in pseudolabel reliability. To

Received 27 November 2025; revised 15 February 2026; accepted 10 March 2026. Date of publication 16 March 2026; date of current version 31 March 2026. This work was supported in part by the National Key Research and Development Program of China under Grant 2024YFE0111800 and Grant 2024YFC3015601 and in part by the National Natural Science Foundation of China under Grant U25A20403. *(Corresponding author: Xuefei Li.)*

The authors are with the National Engineering Research Center for Multimedia Software, School of Computer Science, Wuhan University, Wuhan 430072, China (e-mail: lvliang@whu.edu.cn; qinzilin@whu.edu.cn; snartsnowfly@whu.edu.cn; zhanglefei@whu.edu.cn).

Digital Object Identifier 10.1109/TGRS.2026.3674069

1558-0644 © 2026 IEEE. All rights reserved, including rights for text and data mining, and training of artificial intelligence and similar technologies. Personal use is permitted, but republication/redistribution requires IEEE permission.
See https://www.ieee.org/publications/rights/index.html for more information.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 05,2026 at 19:00:19 UTC from IEEE Xplore. Restrictions apply.

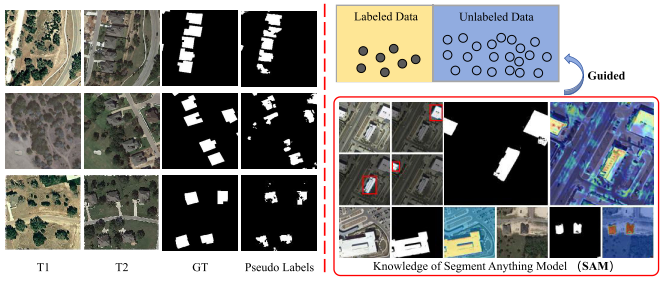


Fig. 1. (a) Difference between the ground-truth label and the pseudolabels generated by the consistency learning framework. Poor-quality pseudolabels are not beneficial for training. (b) Main motivation for introducing SAM, leveraging its extensive knowledge to guide the unsupervised training process.

address this limitation, we propose segment anything model-driven consistency regulation (SAM-CR), a novel SAM-driven consistency regulation framework for SSCD, which enhances pseudolabel credibility by leveraging the extensive visual prior knowledge embedded in foundation models. As shown in Fig. 1(b), the vast generalization capability of the segment anything model (SAM) [22] enables it to serve as a reliable pseudolabel enhancer, particularly under conditions of insufficient manual annotations.

In the proposed framework, inspired by SAM's interactive threshold-based segmentation, we first remove deceptive instances from the pseudolabels generated by the weakly augmented branch to suppress noise. Next, a multimask hybrid hinting strategy is designed to convert the filtered pseudolabels into informative cues for specific changed objects, where a bounding box and a corresponding change center point are generated for each instance. This innovative method transfers remote sensing CD knowledge to SAM without fine-tuning it, which expands the high-confidence regions and reshapes true change instances. Finally, the framework aligns and fuses the pseudolabels enhanced by SAM with the pseudolabels generated by the model to produce robust and trustworthy labels for training, effectively alleviating error accumulation and bias. Furthermore, to exploit the pseudolabels enhanced by SAM more effectively, we also design the multiscale channel similarity (MSCS) strategy within the instance feature alignment (IA) module to enforce feature consistency across perturbed instances. The MSCS strategy leverages the shared semantic information among feature channels of changed objects to construct a multiscale similarity graph, which facilitates the identification and retention of highly correlated channels. This approach effectively suppresses channel noise and provides a promising pathway for achieving fine-grained feature alignment of change instances.

In summary, our contributions are as follows.

1) We propose SAM-CR, an innovative SSCD framework that integrates a strong vision foundation model in consistency learning during training. The experimental results in three public datasets demonstrate the effectiveness of our method.
2) We introduce SAM in the proposed framework and design the multimask hybrid hint strategy to reshape the changed instance and generate more credible pseudolabels in the unsupervised branch.
3) To better leverage the SAM-enhanced pseudolabels, a MSCS strategy is utilized to filter out high-similarity channels to align perturbed objects.

The remainder of this article is organized as follows. Section II introduces the related works. Section III describes the details of SAM-CR. Section IV presents the experimental results of SAM-CR on different datasets. Finally, Section V summarizes this entire article.

II. Related Work

A. Fully Supervised CD

DL methods have become a major research focus in the field of remote sensing CD [23], [24], [25]. As a dense prediction task, CD generally involves inputting bitemporal images, extracting features, performing feature fusion, and ultimately generating a change map. Existing CD frameworks are largely categorized into CNN-based and Transformer-based approaches. Daudt et al. [26] proposed a fully convolutional network that fuses features from bitemporal images to extract change information. The model consists of two subnetworks with identical architectures and shared weights, enabling the extraction of consistent feature representations from both input images within the same distribution domain. BIT [27] introduced a Transformer-based CD method that represents high-level change semantics using visual words and models contextual relationships within a compact, label-guided spatiotemporal space.

In the past few years, a number of advanced CD networks have been proposed. The primary challenges of CD in very high resolution (VHR) remote sensing imagery include mitigating spatial misalignment and distinguishing true semantic changes caused by seasonal variations. The introduction of MetaChanger [28] has drawn increasing attention to the feature interaction stage within CD networks, highlighting the importance of interactive feature extraction for strengthening the model's understanding of the changes of interest. M-Swin [29] is designed to detect subtle changes in small farmland structures by employing hierarchical windowing to capture multiscale information and produce richer boundary details. AANet [30] improves the perception of ambiguous regions by refining convolutional features to generate differential representations and performing adaptive interactive fusion. ELGC-Net [31] efficiently aggregates both global and local contextual information while maintaining a lightweight architecture and employs transposed attention to further minimize computational cost.

However, fully supervised CD relies heavily on large volumes of high-quality annotated data, making it both labor-intensive and resource-demanding. This dependency significantly constrains its scalability and limits its practical deployment in rapidly evolving real-world scenarios.

B. Semisupervised CD

Compared with fully supervised approaches, SSCD significantly reduces the dependence on labeled data, which is one

Authorized licensed use limited to: Hohai University Library. Downloaded on November 05,2026 at 19:00:19 UTC from IEEE Xplore. Restrictions apply.

of its major advantages. In recent years, a variety of learning strategies for leveraging unlabeled data have been explored in SSCD, including adversarial methods, consistency regularization methods, and hybrid approaches. SemiCDNet [32], built upon generative adversarial networks (GANs), employs two discriminators to enhance the consistency of segmentation outputs and entropy distributions between labeled and unlabeled samples. Beyond adversarial strategies, the application of FixMatch [18] to SSCD represents another important contribution and has been widely adopted as the baseline in many studies. FixMatch enforces consistency between weakly and strongly augmented inputs and retains only high-confidence regions from the pseudolabels generated under weak augmentation. This leads to more accurate pseudosupervision and simplifies the overall training process.

Recently, semisupervised semantic segmentation methods in the remote sensing field have achieved remarkable progress [33], [34], and SSCD has developed rapidly. To address rotational nonequivariance and class imbalance, ST-RCL [35] introduces consistency regularization on rotated unlabeled samples and incorporates uncertainty-based class weighting to enhance model generalization. RISL [36] focuses on assessing the reliability of instances to generate more trustworthy pseudolabels, while simultaneously mining rich semantic information from unlabeled data from multiple perspectives. SemiCD-VL [37], a visual–language model (VLM)-based SSCD framework, leverages VLMs to synthesize diverse labels that provide additional supervisory signals for unlabeled samples. AdaSemiCD [20] adopts an active pasting strategy to replace low-confidence regions in pseudolabels, dynamically improving their quality and mitigating sample noise, thereby reducing the bias introduced by low-quality pseudosupervision during training. LGTC [21] proposes a tri-level consistency learning strategy that integrates multilevel supervision at the pixel, region, and image levels, thereby enhancing the model's ability to learn discriminative change features from unlabeled data.

While many advanced methods have emerged, improving the reliability of pseudolabels from unlabeled samples still remains one of the major challenges in the SSCD field.

C. Segmentation Anything Model

The SAM [22] is a milestone in image segmentation, enabling truly practical, general-purpose segmentation across diverse scenarios. Its architecture comprises three core components: a vision encoder, a prompt encoder, and a mask decoder. The vision encoder extracts global and local image features, the prompt encoder processes user-provided hints such as points or bounding boxes, and the mask decoder generates the final segmentation masks. Building on SAM, SAM2 [38] is introduced as a foundational model for both image and video segmentation. In image segmentation tasks, SAM2 delivers improved accuracy and faster inference, representing a significant advancement over the original SAM.

SAM's versatility has enabled its rapid adoption across various DL domains. In remote sensing image segmentation, Qiao et al. [39] preserve SAM's strong boundary-detection capabilities and integrate them with semantic information derived from semantic segmentation models to generate high-quality masks. Ma et al. [40] further enhance SAM-based segmentation by introducing object consistency loss and boundary preservation loss, both computed from SAM's raw outputs, to facilitate more effective model training. SAM has also been leveraged in CD tasks. SAM-CD [41] employs FastSAM [42] as its visual encoder to extract discriminative visual representations from remote sensing imagery, demonstrating strong sample-efficient learning performance. SFCD-Net [43] decomposes SAM's pretrained weights into amplitude and orientation components and fine-tunes its backbone to acquire remote-sensing-specific domain knowledge.

Considering SAM's strong generalization capability, we posit that it can serve as a reliable pseudolabel enhancer within consistency learning frameworks for SSCD.

III. Methodology

In this section, the details of the proposed SAM-CR framework are introduced. Section III-A presents the preliminaries of the consistency learning method in SSCD. Section III-B describes the designed multimask hybrid hint strategy and how to enhance the credibility of pseudolabels with SAM. Section III-C introduces the MSCS strategy in the IA module in detail. The overall framework shown in Fig. 2 is the training procedure of this method.

A. Preliminaries

Given a small set of labeled dataset $\{\mathbf{x}_{ai}^l, \mathbf{x}_{bi}^l, \hat{\mathbf{y}}_i^l\}_{i=1}^{N_l}$ and a large set of unlabeled dataset $\{\mathbf{x}_{ai}^u, \mathbf{x}_{bi}^u\}_{i=1}^{N_u}$, where $\mathbf{x}_i^l$ represents the ith labeled input image and $\mathbf{x}_i^u$ represents the ith unlabeled input image. $\hat{\mathbf{y}}_i^l$ is the ith corresponding label of the labeled dataset. N^l and N^u denote the number of labeled and unlabeled images, and $N^u \gg N^l$.

In the supervised branch, a pair of labeled images is sent to the model, obtaining the probability map to calculate the loss $\mathcal{L}_s$ with the corresponding label, i.e.,

$$\mathcal{L}_s = \frac{1}{B_l}\mathcal{L}_{CE}\left(f_\theta\left(\mathbf{x}_a^l, \mathbf{x}_b^l\right), \hat{\mathbf{y}}^l\right) \tag{1}$$

where $\mathcal{L}_{CE}$ is the cross-entropy loss function, f_θ is the model with the trainable parameters, and B_l indicates the number of labeled images. In the unsupervised branch, based on the weak-to-strong consistency learning method proposed by FixMatch [18], we apply weak data augmentations and strong data augmentations to the unlabeled images to get two pairs of images, weak augmented branch $\{\mathbf{x}_a^w, \mathbf{x}_b^w\}$ and strong augmented branch $\{\mathbf{x}_a^s, \mathbf{x}_b^s\}$. Then the different view pairs are sent to the shared model to get the consistency loss $\mathcal{L}_u$, i.e.,

$$\mathcal{L}_u = \frac{1}{B_u}\mathcal{L}_{CE}\sum \mathbb{1}\left(\max\left(\mathbf{y}^w\right) \geq \tau\right)\left(f_\theta\left(\mathbf{x}_a^s, \mathbf{x}_b^s\right), \hat{\mathbf{y}}^w\right) \tag{2}$$

where B_u indicates the number of unlabeled images, τ is a predefined confidence threshold, $\mathbb{1}$ is an indicator function, and $\hat{\mathbf{y}}^w$ represents the hard label of $\mathbf{y}^w$. $\mathbf{y}^w$ is the probability map from the weak branch, i.e., $\mathbf{y}^w = f_\theta(\mathcal{A}^w(\mathbf{x}_a^w, \mathbf{x}_b^w))$, $\mathcal{A}^w$ denotes strong perturbation. In the end, we get the total

Authorized licensed use limited to: Hohai University Library. Downloaded on November 05,2026 at 19:00:19 UTC from IEEE Xplore. Restrictions apply.

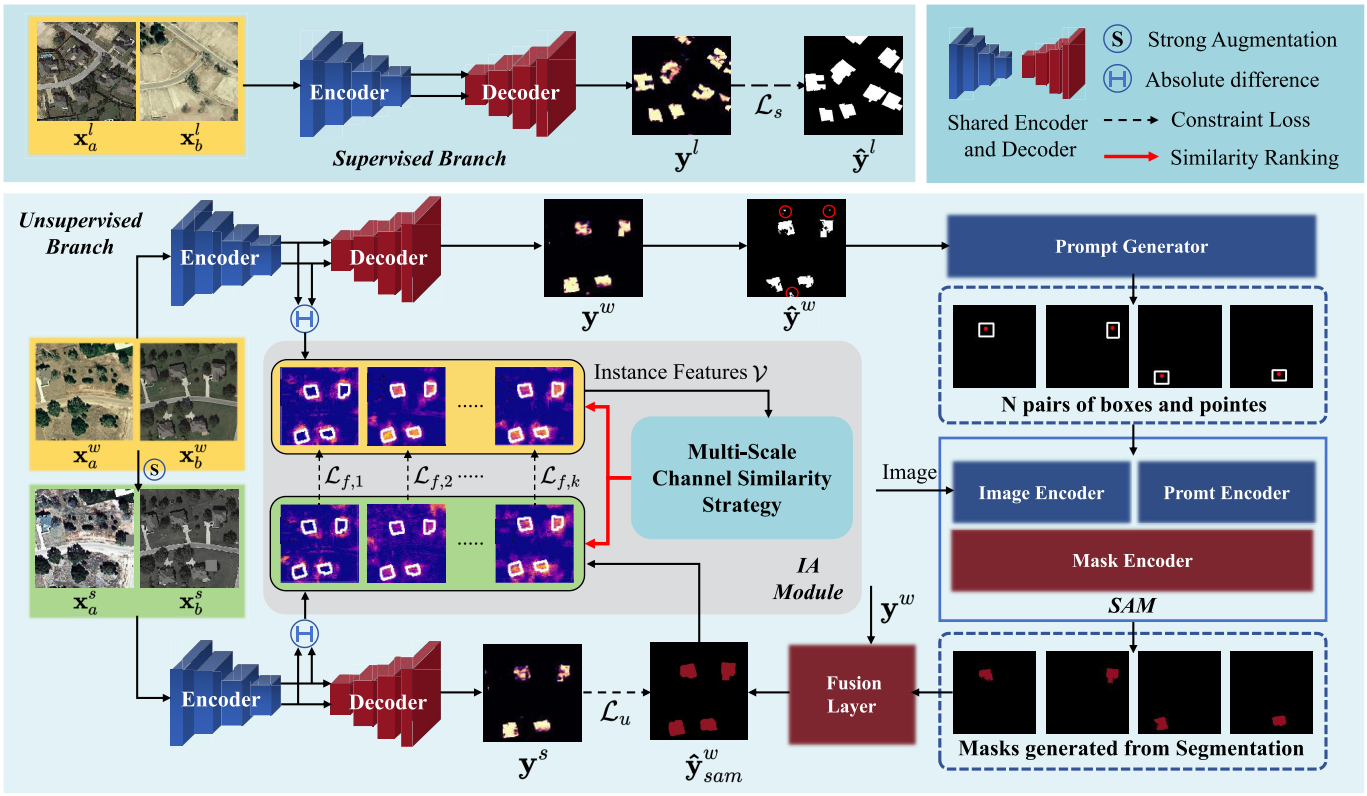


Fig. 2. Illustration of the overall framework of the proposed SAM-CR. The IA module is the IA module, and the red circle in the $\hat{\mathbf{y}}^w$ means that the instance is to be removed.

loss $\mathcal{L}$ in consistency learning from both the supervised and unsupervised branches, i.e.,

$$\mathcal{L} = \frac{1}{2}\left(\mathcal{L}_s + \mathcal{L}_u\right). \tag{3}$$

B. Improving Pseudolabels With SAM

In the consistency learning method of SSCD, improving the reliability of pseudolabels remains a key research focus. Low-quality pseudolabels can easily accumulate prediction biases and propagate errors during training, ultimately leading to suboptimal results. To overcome this problem, we propose a novel approach that integrates SAM into the consistency learning framework, leveraging its powerful segmentation capabilities to enhance the accuracy of changed object regions in pseudolabels. Thanks to its extensive knowledge base, SAM serves as a reliable pseudolabel enhancer, offering a promising solution to the challenge of insufficient annotations. Furthermore, our method removes obviously misleading instances and introduces a multimask hybrid prompt strategy, which incorporates domain-specific knowledge of CD in remote sensing into SAM without requiring fine-tuning. The SAM-driven consistency framework not only refines pseudolabels but also strengthens the model architecture. The details of the approach are described below.

We first obtain the set of changed objects in the pseudolabel, that is, the set of pixels in the foreground area, i.e., $\mathcal{C} = \{\mathbf{C}_1, \mathbf{C}_2, \mathbf{C}_3, \ldots, \mathbf{C}_n\}$. Then, the deceptive objects are

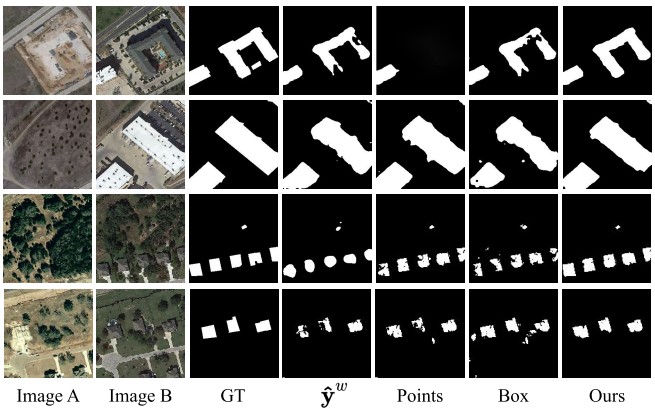


Fig. 3. Pseudolabels generated by SAM based on different prompts, and ours is the designed multimask hybrid hint strategy.

filtered out if the number of the object's boundary pixel set is extremely small. The filtering process is formulated as follows:

$$\mathcal{C}_f = \{\mathbf{C}_i \mid \mathbf{C}_i \in \mathcal{C} \cap \text{contour}\,(\mathbf{C}_i) \geq T\} \tag{4}$$

where the contour function is used to acquire the number of edge pixels of every foreground area, and T is the hyperparameter. This procedure effectively prevents noise interference and eliminates some unnecessary obstacles for the subsequent SAM segmentation. Thus, the length of $\mathcal{C}_f$ set equals the number of connected components in the pseudolabel masks after filtering out noise regions. After acquiring the filtered set of changed objects $\mathcal{C}_f = \{\mathbf{C}_1, \ldots, \mathbf{C}_k\}$, the framework

Authorized licensed use limited to: Hohai University Library. Downloaded on November 05,2026 at 19:00:19 UTC from IEEE Xplore. Restrictions apply.

then transforms them into relevant prompts to guide SAM to locate and reshape the changed objects. In segmentation tasks, SAM typically relies on prompts such as points, boxes, and scribbles. However, as illustrated in Fig. 3, box prompts alone are often insufficient. For instance, when an object exhibits an irregular shape, a bounding box may encompass an overly large area, providing limited spatial guidance for accurate localization. Similarly, as shown in the first row of Fig. 3, point prompts alone may lose substantial contextual information. Considering the diverse orientations and complex geometries of remote sensing objects, relying on a single prompt to refine pseudolabels in SSCD is inadequate. To address this limitation, we propose a novel multimask hybrid prompt strategy that generates both box-level and point-level prompts for each changed object in the pseudolabels. The method first extracts the minimum bounding rectangle of $\mathbf{C}_i$, as the hint box, Box_i

$$Box_i = \mathbf{C}_i \left[x_{\min}, y_{\min}, x_{\max}, y_{\max},\right], \quad C_i \in \mathcal{C}_f \tag{5}$$

$(x_{\min}, y_{\min})$ and $(x_{\max}, y_{\max})$ are the coordinates of the top-left and bottom-right vertices of the generated matrix. Next, we calculate the center points of multiple changed objects $\mathcal{C}_f$. If the geometric center point $p_i(x_i, y_i)$ of each Ci is inside the changed instance, it is selected as the cue point. Conversely, if the p_i falls outside the changed instance, another cue point p_i' will be found. Based on the rotation angle θ of the bounding rectangle, the rotation vector is presented as $(\cos\theta, \sin\theta)$. The search is conducted by increasing the search step in the positive and negative directions of the principal axis of the minimum bounding rectangle until a point is found that falls within the changed instance, i.e.,

$$p_i'\left(x_i + \sigma \times s \times \cos\theta, y_i + \sigma \times s \times \sin\theta\right), \ \sigma \in \{-1, 1\} \tag{6}$$

where σ represents the direction, the parameter s is the search step, and the max value is set to 20. Finally, the cue point $Point_i$ is obtained via

$$Point_i = \{p_i, \mathcal{M}(p_i) = 1\} \cup \left\{p_i', \mathcal{M}\left(p_i'\right) = 0\right\} \tag{7}$$

where $\mathcal{M}$ is the initially obtained pseudolabel. When inputting the prompt box, prompt center point, and the image into SAM, it reshapes the changed instance and then obtains a new $Mask$ map and the corresponding $logits$ map, i.e.,

$$Mask_i, Logits_i = SAM\left(Image, Box_i, Point_i\right), \ i \in [1, k]. \tag{8}$$

There are three strategies for selecting the input $Image$: the first is detecting removed targets from the T1 image, the second is detecting newly appearing targets from the T2 image, and the third is detecting both removed and newly added targets from a new image generated from the pixel-wise differences between T1 and T2. Then, the effective information has been obtained from SAM to correct the potential errors in the pseudolabels, thereby obtaining more reliable corrected pseudolabels. This step is exactly the role of the fusion layer in the Fig. 2, i.e.,

$$\mathbf{y}_{sam}^{w} = \mathbf{y}^{w} + \frac{1}{2}\sum_{i=0}^{k}\left(\mathcal{S}\left(Logits_i\right) - \mathbf{y}^{w}\right) \times Mask_i \tag{9}$$

where $\mathcal{S}$ is the Sigmoid function, and $\mathbf{y}_{sam}^{w}$ represents the new probability map from the weak branch. We finally get the enhanced pseudolabel $\hat{\mathbf{y}}_{sam}^{w}$, which is the hard label of y_{sam}^{w}. Then, the difference is calculated between the enhanced pseudolabels by SAM and the prediction map of the strong enhancement branch to get the cross-entropy loss $\mathcal{L}_u$.

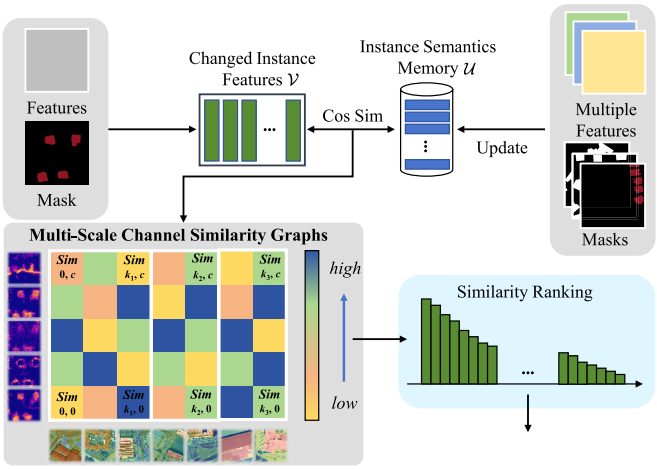


Fig. 4. Changed instance features are extracted from the difference feature maps and pseudolabels. The instance semantic memory is derived from the training data of labeled and unlabeled streams. The "Cos Sim" represents the cosine similarity function. The MSCS graph is calculated and then sorted according to similarity to filter for highly similar channels.

C. MSCS Strategy

To fully leverage the pseudolabels enhanced by SAM, we introduce an IA module that enforces the alignment of perturbed change instances in the feature space prior to feature fusion. Humans tend to perceive an object as a coherent whole when observing its attributes, which aligns with the inherent nature of CD tasks. Inspired by this, we propose an MSCS strategy to generate channel similarity maps between corresponding change instances. This strategy identifies channels with high similarity to represent the changed instances highlighted in the difference feature map, indicating the most promising directions for alignment. Moreover, the limited number of real change instances in a small batch is often insufficient to provide effective guidance, necessitating richer feature representations for channel selection. To address this problem, we introduce an instance semantics memory to store channel information. The MSCS strategy equips the model with a deeper understanding of changed objects, including texture and color, thereby complementing the limitations of consistency learning during feature extraction. The module is described in detail as follows.

As shown in Fig. 4, we get the feature difference maps $\mathbf{F}^{w}$ and $\mathbf{F}^{s}$ from the weak and strong branches, and obtain the relevant changed instance feature $\mathcal{V} = \{\mathbf{v}_1, \mathbf{v}_2, \ldots, \mathbf{v}_n\}$ by the foreground pixel position of the enhanced pseudolabel $\hat{\mathbf{y}}_{sam}^{w}$. And the stored instance features $\mathcal{U} = \{\mathbf{u}_1, \mathbf{u}_2, \ldots, \mathbf{u}_m\}$ are retrieved from the instance semantics memory. Because most of the objects to be detected in existing CD datasets are artificial creations, which often have similar physical characteristics, we use the channel similarity of the changed objects to screen out highly similar channels and achieve more

Authorized licensed use limited to: Hohai University Library. Downloaded on November 05,2026 at 19:00:19 UTC from IEEE Xplore. Restrictions apply.

robust consistency alignment. The channel mean value $\bar{\mathbf{v}}$, $\bar{\mathbf{u}}$ of instances of the same class is obtained via

$$\bar{\mathbf{v}} = \frac{1}{n}\sum_{i=1}^{n}\frac{\mathbf{v_i}}{\|\mathbf{v_i}\|_2}, \quad \bar{\mathbf{u}} = \frac{1}{m}\sum_{i=1}^{m}\frac{\mathbf{u_i}}{\|\mathbf{u_i}\|_2}. \tag{10}$$

Then, the cosine similarity matrix can be obtained using the above channel vectors. Similarity estimated by a small number of neighbors tends to focus on the local region, while graphs with many neighbors lead to a smoother estimation, providing a larger view. To obtain robust similarity estimation, the MSCS strategy is designed, i.e.,

$$S_{ij} = \mathrm{TopK}\left(\cos\left(\frac{\bar{\mathbf{v}}}{\|\bar{\mathbf{v}}\|_2}, \frac{\bar{\mathbf{u}}}{\|\bar{\mathbf{u}}\|_2}\right), w_i\right), \ w_i \in (w_1, w_2, w_3). \tag{11}$$

The average value of MSCS is calculated, i.e.,

$$S_{ins} = \frac{1}{3}\sum_{i=1}^{3}\sum_{j=1}^{w_i}\frac{S_{ij}}{w_i}. \tag{12}$$

Based on the final channel similarity value, we use the same TopK function to filter out the K channels $\mathcal{Q} = \{Seq_1, Seq_2, \ldots, Seq_k\}$ with the highest similarity to represent the changed instance, thus minimizing the interference of noise in the channel.

Based on the specific channels obtained, the loss of IA can be calculated between strong and weak branches, i.e.,

$$\mathcal{L}_f = \frac{1}{K}\sum_{i=1}^{K}\mathcal{L}_{MSE}\left(\mathbf{F}^s_{c_i}, \mathbf{F}^w_{c_i}\right) \times \hat{\mathbf{y}}^w_{sam}, \quad c_i \in \mathcal{Q} \tag{13}$$

where K is the length of $\mathcal{Q}$ and $\hat{\mathbf{y}}^w_{sam}$ is the obtained hard label enhanced by SAM.

Finally, the overall loss of our proposed framework is the combination of the supervised loss $\mathcal{L}_s$, the consistency learning loss $\mathcal{L}_u$, and the changed feature alignment loss $\mathcal{L}_f$, i.e.,

$$\mathcal{L} = \mathcal{L}_s + \lambda_1\mathcal{L}_u + \lambda_2\mathcal{L}_f \tag{14}$$

where λ_1 and λ_2 denotes the weights of the loss terms.

IV. Experiment Results

A. Dataset

LEVIR-CD [44] Dataset: The VHR Architectural CD dataset, provided by Beihang University, consists of 637 pairs of high-resolution image patches captured between 2002 and 2018. The image size is 1024 × 1024 pixels, with a spatial resolution of 0.5 m. This dataset focuses on architectural variations, including residential villas and high-rise apartments, from 20 different areas in several cities in Texas, including Pflugerville, TX, Lakeway, Kyle, Austin, Dripping Springs, Buda, and Lakeway. The default image pairs are cropped into nonoverlapping patches of 256 × 256.

WHU-CD [45] Dataset: The WHU-CD dataset is a large-scale VHR dataset for building CD, publicly provided by Wuhan University. It consists of two pairs of aerial images taken in Christchurch, New Zealand, in 2012 and 2016. The entire image pair measures 32 507 × 15 354 pixels, with a pixel resolution of 0.075 m. Following the common split, the dataset is divided into 80% training, 10% validation, and the remaining 10% testing. For ease of comparison, the images are also cropped into 256 × 256 blocks with no overlap, resulting in 5947, 743, and 744 pairs of blocks for training, validation, and testing, respectively.

CDD-CD [46] Dataset: The CDD dataset is cropped from seasonal remote sensing imagery from Google Earth, with spatial resolutions ranging from 3 to 100 cm/px. The dataset contains 16 000 image pairs, with 10 000, 3000, and 3000 used for training, validation, and testing, respectively. These images are all 256 × 256 pixels and contain many changed objects of many classes, such as buildings, roads, and vegetation.

B. Implementation Detail

In this experiment, due to GPU memory and computing power limitations, we set the batch size and epoch number to 8 and 50, respectively, after numerous attempts, to achieve model convergence. For the strong augmentation branch, we primarily adopted RandAugment [47] to augment the unlabeled images, applying transformations such as Color, Autocontrast, Equalize, Identity, Contrast, Brightness, Sharpness, Posterize, and Solarize. During the training phase, the loss function is minimized by the AdamW optimizer with a weight decay of 0.05 and a learning rate of $1e^{-4}$. The proposed model was implemented in PyTorch and trained, validated, and tested on a single NVIDIA RTX 3090 GPU. After each training epoch, evaluation was performed on the validation set, and the model achieving the highest $F1$ score was saved as the final best-performing model.

C. Evaluation Metrics

To evaluate the performance of the SAM-CR model, we introduce five CD evaluation metrics: overall accuracy (OA), precision (Pre.), recall (Rec.), intersection over union (IOU), and $F1$-score ($F1$), and the formulas are listed below

$$\mathrm{OA} = \frac{\mathrm{TP}+\mathrm{TN}}{\mathrm{TP}+\mathrm{TN}+\mathrm{FN}+\mathrm{FP}} \tag{15}$$

$$\mathrm{Pre.} = \frac{\mathrm{TP}}{\mathrm{TP}+\mathrm{FP}} \tag{16}$$

$$\mathrm{Rec.} = \frac{\mathrm{TP}}{\mathrm{TP}+\mathrm{FN}} \tag{17}$$

$$\mathrm{IoU} = \frac{\mathrm{TP}}{\mathrm{TP}+\mathrm{FN}+\mathrm{FP}} \tag{18}$$

$$F1 = \frac{2\times\mathrm{Pre.}\times\mathrm{Rec.}}{\mathrm{Pre.}+\mathrm{Rec.}} \tag{19}$$

where TN denotes the number of correctly classified unchanged pixels, TP denotes the number of correctly classified changed pixels, FN denotes the number of wrongly classified unchanged pixels, and FP denotes the number of wrongly classified changed pixels. The five evaluation metrics range from 0% to 100%.

D. Comparison Experiments

To demonstrate the effectiveness and robustness of the proposed SAM-CR framework, the comparative methods discussed below include s4GAN [48], AdvEnt [49], FixMatch [18], CPS [50], RCR [51], ST-RCL [35], ECPS [52], and

Authorized licensed use limited to: Hohai University Library. Downloaded on November 05,2026 at 19:00:19 UTC from IEEE Xplore. Restrictions apply.

TABLE I

QUANTITATIVE RESULTS OF DIFFERENT METHODS ON THE LEVIR-CD DATASET WITH DIFFERENT LABELED RATIOS. ALL SCORES ARE EXPRESSED IN PERCENTAGE (%)

Methods	5%					10%					20%					40%				
	OA	Pre	Rec	F1	IoU	OA	Pre	Rec	F1	IoU	OA	Pre	Rec	F1	IoU	OA	Pre	Rec	F1	IoU
s4GAN [48]	97.84	87.22	63.61	73.57	58.19	97.97	88.25	65.74	75.35	60.45	98.10	86.33	70.83	77.82	63.69	98.41	85.03	80.57	82.74	70.56
Advent [49]	98.34	85.18	78.51	81.71	69.08	98.58	86.89	82.33	84.55	73.23	98.74	89.17	83.32	86.14	75.66	98.77	88.51	84.90	86.67	76.47
CPS [50]	98.67	91.32	80.98	85.19	74.18	98.83	91.19	82.79	87.18	77.30	98.77	90.17	86.19	88.37	79.17	98.96	91.14	86.07	88.35	80.24
RCR [51]	98.79	89.82	83.84	86.72	76.56	98.85	89.92	85.18	87.49	77.76	98.82	87.54	87.40	87.47	77.73	98.86	87.99	87.93	87.96	78.51
FixMatch [18]	98.80	**93.37**	81.62	86.89	76.72	98.82	**95.24**	80.69	87.10	76.61	98.88	**94.19**	82.59	87.98	78.68	98.87	89.84	86.99	87.74	79.97
ST-RCL [35]	-	88.75	**86.67**	87.70	78.09	-	91.42	86.23	88.75	79.77	-	91.06	87.91	89.46	80.93	-	91.49	88.11	89.77	81.43
ECPS [52]	98.74	88.88	83.78	86.25	75.83	98.80	90.07	83.77	86.81	76.68	98.86	92.32	82.68	87.23	77.36	98.94	**91.98**	84.93	88.32	79.08
LGTC [21]	98.83	90.55	83.94	87.12	77.18	98.90	89.24	**87.07**	88.14	78.80	98.99	90.85	87.39	89.09	80.32	99.00	91.30	87.13	89.17	80.46
Ours	**98.93**	91.15	86.30	**88.63**	**79.52**	**98.97**	93.34	85.45	**89.24**	**80.81**	**99.08**	92.23	**88.70**	**90.39**	**82.25**	**99.08**	91.42	**89.74**	**90.58**	**82.77**

TABLE II

QUANTITATIVE RESULTS OF DIFFERENT METHODS ON THE WHU-CD DATASET WITH DIFFERENT LABELED RATIOS. ALL SCORES ARE EXPRESSED IN PERCENTAGE (%)

Methods	5%					10%					20%					40%				
	OA	Pre	Rec	F1	IoU	OA	Pre	Rec	F1	IoU	OA	Pre	Rec	F1	IoU	OA	Pre	Rec	F1	IoU
s4GAN [48]	98.16	88.80	61.45	72.63	57.03	98.49	84.74	75.57	79.89	66.52	98.69	87.98	77.65	82.49	70.20	98.90	90.63	80.63	85.34	74.43
Advent [49]	98.68	88.47	76.86	82.26	69.86	98.76	84.83	83.82	84.32	72.89	99.02	91.61	82.86	87.02	77.02	99.22	90.67	89.55	90.11	82.00
CPS [50]	98.94	89.96	81.58	85.84	75.51	99.06	87.07	87.59	87.57	78.41	99.20	91.50	88.28	89.91	81.69	99.29	93.56	89.53	91.27	83.72
RCR [51]	98.87	89.98	80.48	84.97	73.86	99.13	92.41	85.16	88.64	79.60	99.07	89.67	86.55	88.08	78.70	99.31	90.90	**91.78**	91.33	83.65
FixMatch [18]	99.07	92.32	83.38	87.64	78.04	99.14	94.10	83.55	88.51	79.39	99.23	**94.25**	85.72	89.78	81.46	99.27	**93.89**	87.58	90.76	83.20
ST-RCL [35]	-	92.20	83.80	87.80	78.25	-	91.45	84.79	88.00	78.57	-	91.90	86.82	89.29	80.65	-	92.93	89.56	91.21	83.84
ECPS [52]	99.12	92.29	84.04	88.43	79.19	99.16	91.92	87.11	89.41	80.69	99.19	91.02	88.73	89.81	81.26	99.37	92.41	91.63	92.01	85.24
LGTC [21]	99.12	90.85	**86.55**	88.65	79.61	99.20	**92.47**	87.04	89.67	81.27	99.26	92.74	88.21	90.42	82.52	99.37	92.37	91.70	92.04	85.25
Ours	**99.14**	**92.54**	85.32	**88.79**	**79.91**	**99.27**	91.44	**90.04**	**90.88**	**83.21**	**99.30**	93.21	**89.16**	**91.09**	**83.54**	**99.44**	93.87	91.72	**92.89**	**87.08**

LGTC [21]. s4GAN and AdvEnt are adversarial-learning-based approaches: the former approximates the feature space of real samples by leveraging large-scale synthetic visual data, while the latter enhances model performance by combining adversarial loss with standard cross-entropy loss. FixMatch generates pseudolabels for unlabeled images under weak augmentation and uses them to supervise predictions under strong augmentation, retaining only high-confidence regions through strict confidence thresholds. CPS enforces consistency between the predictions of two perturbed networks for the same input image and expands the training data using unlabeled data with pseudolabels. RCR imposes feature-level consistency constraints to improve SSCD performance. ST-RCL further integrates consistency learning on rotated unlabeled data and employs uncertainty-based class weighting in the unsupervised branch. ECPS performs cross-training on unlabeled data and fuses multiple pseudolabels generated by different networks to obtain higher-quality supervision. LGTC enforces alignment at pixel, region, and image levels, providing complementary supervision signals from fine-grained to global semantic scales. In our experiments, four proportions (5%, 10%, 20%, and 40%) of labeled data are used for supervised training, while the remaining samples are treated as unlabeled data for unsupervised learning.

Table I presents the quantitative results of all compared methods on the LEVIR-CD dataset under four supervised labeled ratios. The best values for each performance metric are highlighted in bold. Across all labeled ratios, the proposed framework achieves the highest performance in the three key metrics: IoU, $F1$ score, and OA. Notably, in terms of the $F1$ score, our method outperforms the second-best approach by more than 1%, demonstrating a clear advantage over other comparison methods. For precision and recall, while our method does not always achieve the top score under certain labeled ratios, it consistently ranks among the second or third highest. Moreover, compared to the baseline method FixMatch, the recall metric shows a substantial improvement, indicating that the model effectively reduces missed detections of true change instances across diverse scenes. This is consistent with our initial objective, which is to improve the completeness of the target regions to be detected.

Table II reports the quantitative results on the WHU-CD dataset under the same labeled ratios. Although performance on this dataset is slightly lower than on LEVIR-CD, the proposed method still achieves the highest IoU, $F1$ score, and OA. Specifically, with a 10% labeled ratio, the $F1$ score surpasses the second-best method by 1.21%. When the labeled ratio is 40%, the IoU exceeds 87%, representing a substantial improvement over other advanced methods. Table III summarizes the results on the CDD-CD dataset under the four labeled ratios. Consistent with the findings on the previous two datasets, the proposed framework demonstrates excellent

Authorized licensed use limited to: Hohai University Library. Downloaded on November 05,2026 at 19:00:19 UTC from IEEE Xplore. Restrictions apply.

TABLE III
QUANTITATIVE RESULTS OF DIFFERENT METHODS ON THE CDD-CD DATASET WITH DIFFERENT LABELED RATIOS. ALL SCORES ARE EXPRESSED IN PERCENTAGE (%)

Methods	5%					10%					20%					40%				
	OA	Pre	Rec	F1	IoU	OA	Pre	Rec	F1	IoU	OA	Pre	Rec	F1	IoU	OA	Pre	Rec	F1	IoU
s4GAN [48]	93.84	85.11	57.88	68.90	52.56	94.84	91.06	62.41	74.06	58.80	96.09	91.56	73.66	81.64	68.97	96.51	91.32	77.86	84.05	72.49
Advent [49]	94.91	85.17	73.42	78.86	65.10	96.21	91.90	77.48	84.08	72.53	97.17	91.11	86.33	88.66	79.62	97.63	**93.18**	88.15	90.59	82.81
CPS [50]	94.53	84.56	70.36	76.81	62.35	96.27	92.03	77.86	84.36	72.94	97.30	92.56	86.22	89.27	80.63	97.51	90.64	89.75	90.16	81.89
RCR [51]	94.54	85.07	70.06	76.84	62.39	95.78	88.87	76.99	82.51	70.22	96.70	90.57	83.15	86.70	76.52	97.58	93.11	87.78	90.37	82.43
FixMatch [18]	95.67	89.57	75.68	82.02	69.99	96.88	89.32	85.23	87.34	77.49	97.51	90.27	90.09	90.12	82.04	97.65	90.64	89.81	90.77	83.01
ST-RCL [35]	-	84.32	78.84	81.49	68.76	-	92.10	85.45	88.65	79.62	-	92.56	89.50	91.01	83.50	-	92.70	89.73	91.19	83.81
ECPS [52]	95.81	85.22	78.30	81.67	69.29	96.83	**94.03**	80.56	86.77	76.58	97.39	90.09	88.40	89.31	81.21	97.71	90.74	90.09	91.06	83.45
LGTC [21]	96.01	**90.93**	76.76	83.25	71.30	96.94	93.13	82.43	87.46	77.71	97.52	**93.41**	86.94	90.06	81.92	97.80	93.63	89.08	91.30	83.99
Ours	**96.30**	90.26	**80.78**	**84.16**	**73.11**	**97.36**	91.36	**86.81**	**89.02**	**80.45**	**97.79**	91.51	**90.74**	**91.15**	**83.76**	**97.93**	91.33	**92.19**	**91.81**	**85.03**

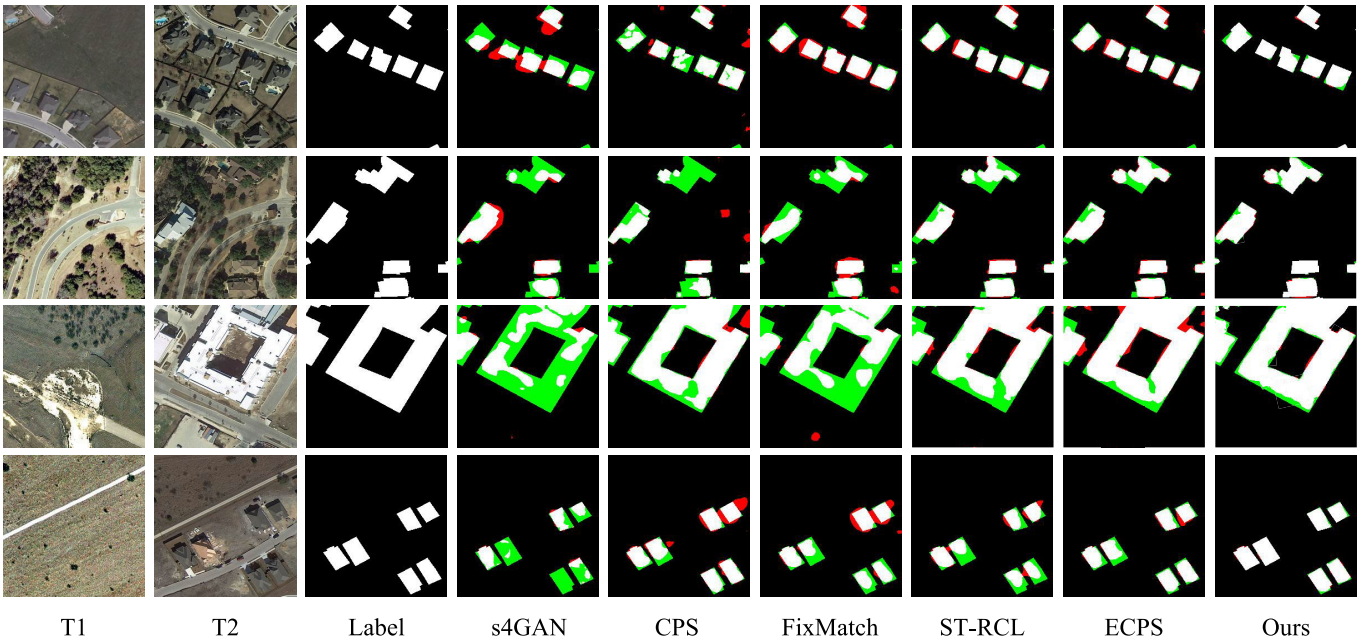


Fig. 5. Comparison of result mask maps by various methods on LEVIR-CD dataset. The red and green colors represent FP and FN results, respectively.

performance, confirming its robustness and effectiveness across multiple CD benchmarks.

To visually illustrate the superiority and robustness of the proposed SAM-CR, Figs. 5–7 present comparisons with other SSCD methods. In these visualizations, white indicates true positives, black indicates true negatives, red indicates false positives, and green indicates false negatives. For each dataset, we select four representative images and show the model inference results at a labeled ratio of 10%. As shown in the figures, compared to the baseline method FixMatch, SAM-CR produces more complete predictions of true change regions while simultaneously reducing some spurious change regions. Notably, SAM-CR achieves more precise and clearer delineation of building edges. This improvement can be attributed to SAM's powerful segmentation capabilities, which reshape change instances during training and provide the model with more accurate guidance for learning and feature exploration.

Similarly, compared to other state-of-the-art SSCD methods, our approach continues to demonstrate clear advantages, consistent with the quantitative results presented in Section IV-D. As shown in the figures, the proposed method occasionally exhibits missed or ambiguous detections, a challenge common to all existing approaches. This limitation can be attributed in part to substantial long-term geographical changes and shadows cast by densely clustered buildings, which hinder the model's ability to accurately identify objects. At the same time, these observations highlight potential areas for further improvement. For instance, leveraging SAM pretrained on large-scale remote sensing imagery in future work could enable our framework to achieve even more accurate and robust results.

E. Effectiveness of Introducing SAM

The primary motivation for integrating SAM into the consistency learning framework for SSCD is to enhance the feasibility and robustness of pseudolabels. Previous studies indicate that high-confidence pseudolabels can mitigate error

Authorized licensed use limited to: Hohai University Library. Downloaded on November 05,2026 at 19:00:19 UTC from IEEE Xplore. Restrictions apply.

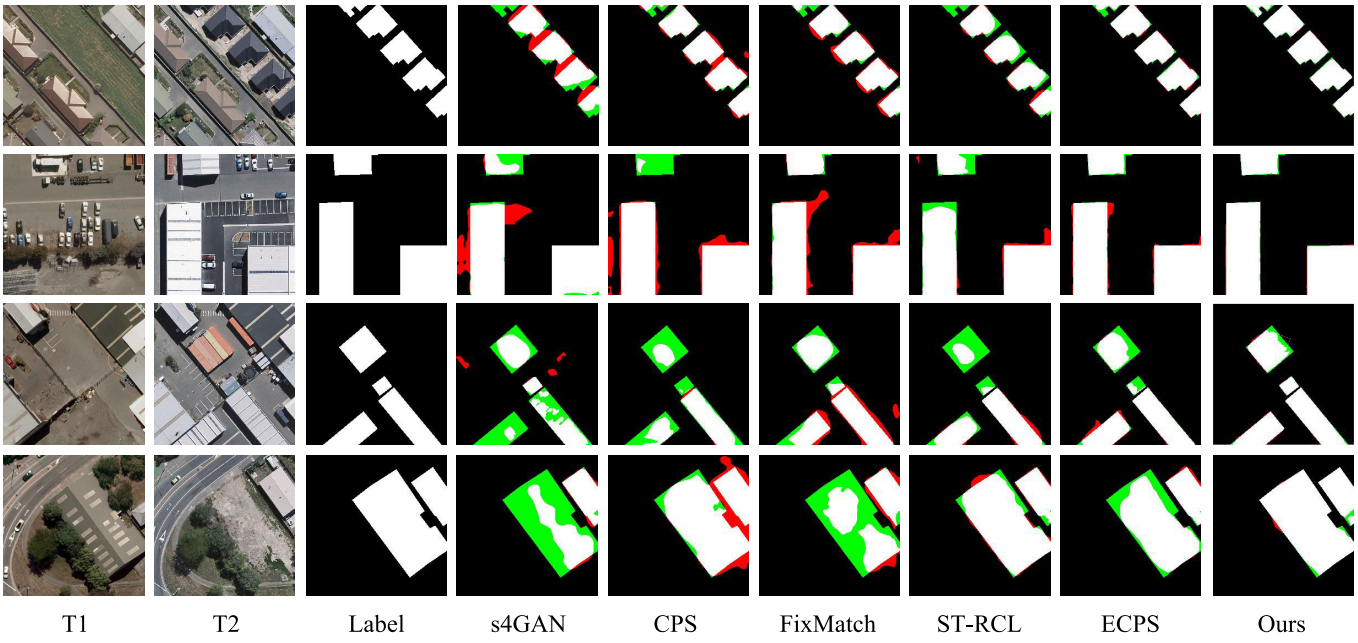


Fig. 6. Comparison of result mask maps by various methods on WHU-CD dataset. The red and green colors represent FP and FN results, respectively.

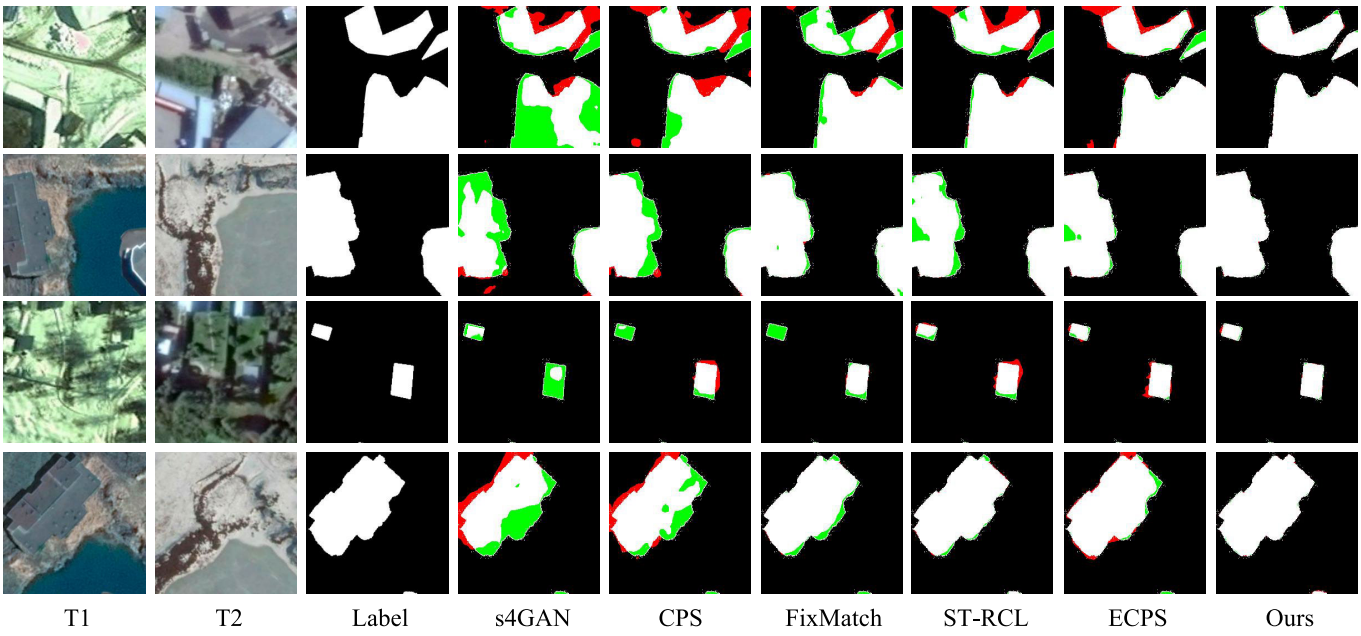


Fig. 7. Comparison of result mask maps by various methods on CDD-CD dataset. The red and green colors represent FP and FN results, respectively.

accumulation and prediction bias during training, thereby providing a safer and more effective supervisory signal. We aim to validate this claim using intermediate model results. Fig. 8(a)–(d) depict the average mIoU between the initial pseudolabels and SAM-enhanced pseudolabels at different training epochs for the LEVIR-CD dataset under various labeled ratios (evaluated solely against ground-truth labels, which are not used during training). The results demonstrate that SAM-enhanced pseudolabels consistently outperform the initial pseudolabels throughout all training stages, indicating their ability to provide higher-quality supervisory signals across epochs. Furthermore, as the proportion of labeled data increases, the proposed model generates virtual labels with more stable accuracy after training for the same number of iterations, whereas the baseline model exhibits significant fluctuations in accuracy.

Table IV presents several representative epochs to illustrate the quantitative differences between the baseline method and our approach. Notably, at the first epoch with a 5% labeled ratio, the mIoU increased by more than 4%. At a marking ratio of 20%, the advantage is somewhat reduced, but the mIoU of the pseudolabel still shows a significant improvement

Authorized licensed use limited to: Hohai University Library. Downloaded on November 05,2026 at 19:00:19 UTC from IEEE Xplore. Restrictions apply.

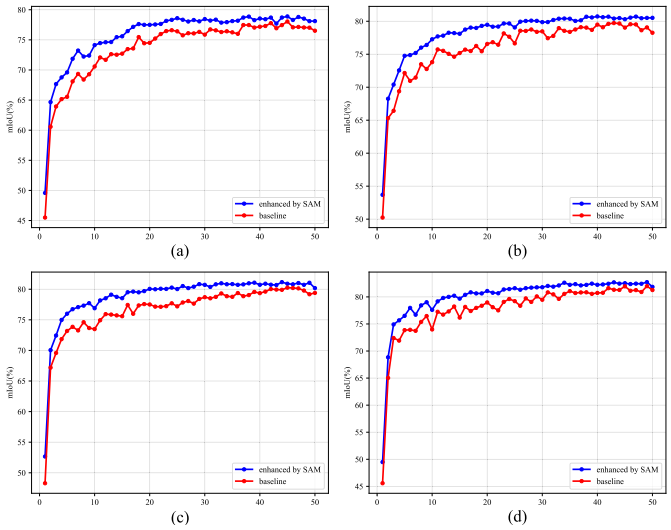


Fig. 8. (a)–(d) Average IoU comparison curve of the generated pseudolabels for each epoch on the LEVIR-CD dataset with a labeled ratio of 5%, 10%, 20%, and 40%, respectively.

TABLE IV
COMPARATIVE RESULTS WITH MIOU (%) METRIC OF PSEUDOLABELS GENERATED BY THE BASELINE METHOD AND OUR METHOD

Epoch	5%			20%		
	Baseline	Ours	△	Baseline	Ours	△
1	45.49	49.58	↑**4.09**	48.29	52.66	↑**4.37**
5	65.54	69.59	↑**4.05**	73.18	76.00	↑**2.82**
10	70.58	74.13	↑**3.55**	73.49	76.90	↑**3.41**
20	74.51	77.50	↑**2.99**	77.51	80.06	↑**2.55**
30	75.87	78.45	↑**2.58**	78.70	80.72	↑**2.02**
40	77.18	78.55	↑**1.37**	79.36	80.73	↑**1.37**
45	78.09	78.89	↑**0.80**	80.26	80.90	↑**0.64**
50	76.52	77.24	↑**0.72**	79.40	80.21	↑**0.81**

TABLE V
ABLATION STUDY OF EVERY COMPONENT OF THE SAM-CR FRAMEWORK ON THE 5% LABELED LEVIR-CD

Baseline	SAM	IDA	IDA+MSCS	F1	IoU	△
√				86.89	76.72	**0.00**
√	√			87.93	78.56	↑**1.84**
√		√		86.93	76.79	↑**0.07**
√			√	87.06	77.03	↑**0.31**
√	√	√		88.07	78.73	↑**2.01**
√	√		√	88.63	79.52	↑**2.80**

compared to the baseline method. Although the gap narrows as training progresses, pseudolabels optimized with SAM remain more reliable than the original labels. These findings underscore the effectiveness of incorporating SAM and our proposed multimask hybrid hinting strategy in the SSCD task.

F. Ablation Study

1) Components of SAM-CR: To further investigate the contributions of each component in SAM-CR, we perform an ablation study in this section. As shown in Table V, six ablation experiments are conducted on the LEVIR-CD dataset using a 5% labeled ratio. Introducing SAM to enhance the pseudolabels in the unsupervised branch increases the IoU score by 1.84%, representing a substantial improvement. This underscores the importance of integrating the powerful SAM segmentation model into the consistency learning framework for SSCD. Interestingly, directly adding the IA module and the MSCS strategy to enforce bitemporal feature alignment without SAM yields only a marginal improvement, with the IoU increasing by just 0.31%. We attribute this limited effect to the low confidence of the pseudolabels, which results in inaccurate localization of changing instances and the accumulation of erroneous information. This limitation is alleviated when SAM is employed to enhance pseudolabel confidence. Moreover, the MSCS strategy leads to a notable improvement, boosting the IoU by 0.79%. By leveraging the fact that most changing objects are artificial structures, this strategy effectively selects informative channels, suppresses potential noise, and strengthens the semantic consistency of the predictions.

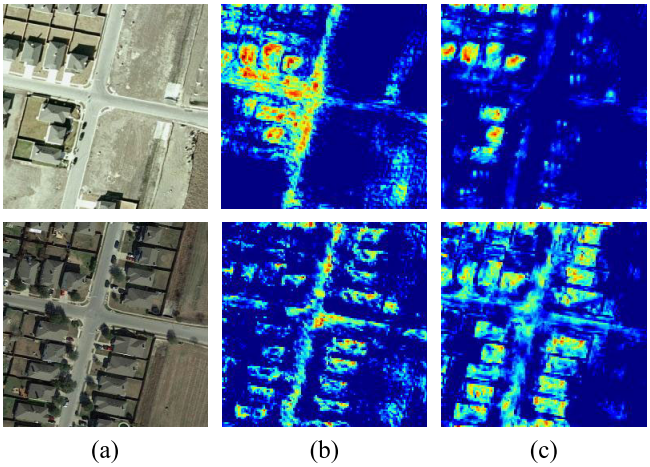


Fig. 9. (a) Original image. (b) and (c) Image feature visualization results, calculated from the encoder outputs. (b) Without the MSCS strategy. (c) With the MSCS strategy.

As shown in Fig. 9, the MSCS strategy within the IA module enables the model to generate more salient feature maps. For instance, in Fig. 9(c), the buildings are located more accurately, with clearer edges and more complete shapes. This improvement facilitates the recognition of changed instances and the generation of robust pseudolabels in the subsequent decoder. The enhancement arises because the MSCS strategy filters out noise prior to alignment, reducing interference from redundant information and allowing the encoder to focus more effectively on the target objects during feature extraction.

In summary, each component of the SAM-CR framework has a positive impact on the overall performance of the model. Moreover, both the introduction of SAM and the IDA module are only activated during training and do not incur additional inference cost.

2) Hyperparameter Settings: In (14), the unsupervised loss includes the consistency loss $\mathcal{L}_u$ and the changed feature alignment loss $\mathcal{L}_f$. As shown in Table VI, we conducted an ablation study on the two hyperparameters λ_1 and λ_2 involved. Both types of losses play an important role in training on unlabeled data and contribute positively to the model's strong performance. However, it is evident from the table that the

Authorized licensed use limited to: Hohai University Library. Downloaded on November 05,2026 at 19:00:19 UTC from IEEE Xplore. Restrictions apply.

TABLE VI
ABLATION STUDY OF WEIGHTS λ_1 AND λ_2 FOR CONSISTENCY LOSS AND ALIGNMENT LOSS

Hyperparameter		5%		20%	
λ_1	λ_2	F1	IoU	F1	IoU
0.1	0.1	88.00	78.36	89.78	80.91
0.1	0.5	88.12	78.61	90.07	81.56
0.5	0.1	88.17	78.69	90.06	81.49
0.5	0.5	**88.63**	**79.52**	**90.39**	**82.25**
0.5	1	87.69	77.61	89.91	81.74
1	1	87.38	76.94	89.84	81.62

TABLE VII
ABLATION STUDY OF THE THRESHOLD OF INSTANCE EDGE PIXELS COUNT T ON THE TWO DATASETS

T	LEVIR-CD				WHU-CD			
	Pre	Rec	F1	IoU	Pre	Rec	F1	IoU
-	89.49	**87.51**	88.23	78.88	89.29	87.61	88.34	79.18
5	**93.38**	83.87	88.37	79.17	89.11	88.02	88.56	79.47
10	91.15	86.30	**88.63**	**79.52**	**92.54**	85.32	**88.79**	**79.91**
15	90.63	85.87	88.19	78.87	92.40	84.71	88.34	79.22
20	90.17	85.75	88.01	78.56	87.12	**88.56**	87.95	78.56

TABLE VIII
ABLATION STUDY OF THE NUMBER OF CHANNELS FILTERED OUT K ON THE 5% LABELED LEVIR-CD

K	Pre	Rec	F1	IoU
16	**92.69**	82.57	87.91	78.47
32	91.79	85.09	88.37	79.01
48	92.16	84.88	88.39	79.04
64	91.15	86.30	**88.63**	**79.52**
128	89.06	**87.46**	88.14	78.68

choice of hyperparameters is also critically important. When $\lambda_1 = 0.5$ and $\lambda_2 = 0.5$, the model's performance reaches the best. When the weight of unsupervised loss is too high, the performance decreases, such as when $\lambda_1 = 1$ and $\lambda_2 = 1$. We attribute this to the fact that pseudolabels inevitably contain errors, even though our method optimizes consistency learning between the strong and weak branches. When the proportion of unsupervised loss is too high, it can misdirect the model's learning process, a challenge commonly observed across all methods.

We conducted an ablation study on the hyperparameter T in (4). As shown in Table VII, the threshold T directly influences the identification of deceptive objects in the initial labels. When T is set too low, residual noise remains in the pseudolabels, hindering SAM's ability to accurately reconstruct true changed objects. Conversely, when T is set too high, true positive samples may be erroneously discarded. Experiments on the LEVIR-CD and WHU-CD datasets with a labeled ratio of 5% indicate that the optimal performance is achieved when $T = 10$. The consistent conclusions across the two datasets can be attributed to their shared focus on building CD, where object sizes tend to be similar across various scenes.

The ablation study on the hyperparameter K in (13) is also performed. As shown in Table VIII, the number of highly similar channels selected by the MSCS strategy directly affects its performance. Experiments on the LEVIR-CD dataset with a labeled ratio of 5% demonstrate that the model obtains optimal results when $K = 64$. Selecting fewer channels may fail to capture the diverse features present in remote sensing images, leading to information loss.

V. CONCLUSION

In this article, we propose a novel consistency learning framework named SAM-CR for the SSCD task. Addressing the critical challenge of improving the credibility and robustness of pseudolabels, we first integrate SAM to enhance pseudolabel quality. Within this framework, SAM serves as a reliable pseudolabel enhancer, leveraging its extensive knowledge base and superior segmentation capabilities. To further guide SAM's segmentation for SSCD, we design a multimask hybrid hint strategy, which achieves excellent performance on remote sensing images without requiring SAM fine-tuning. Both quantitative and qualitative results show that this approach expands the high-confidence regions in pseudolabels, thereby reducing error accumulation in the unsupervised branch during training caused by inaccurate pseudolabels. To better exploit the SAM-enhanced pseudolabels, we introduce an IA module after the encoder and propose a MSCS strategy. This approach suppresses channel noise while enforcing alignment of perturbed change instances, providing safer and more effective supervision. Extensive experiments on three remote sensing CD datasets, including LEVIR-CD, WHU-CD, and CDD-CD, show that SAM-CR outperforms state-of-the-art SSCD methods. The IoU improves by more than 1.5% across nearly all labeled ratios, and other metrics also show strong performance gains. Ablation studies further verify that both the integration of SAM and the MSCS strategy contribute positively to the overall performance.

Nevertheless, SAM-CR still has room for improvement. For example, its segmentation performance degrades when processing a large number of small changed instances, and the computational cost increases accordingly. Therefore, designing simpler and more efficient strategies to enhance the credibility of pseudolabels, or adding boundary-aware and local attention modules to enhance the model's sensitivity to the boundaries of subtle changes, will become an important and valuable research direction in the field of SSCD.

ACKNOWLEDGMENT

The numerical calculations in this work were supported by the supercomputing system in the Supercomputing Center of Wuhan University.

REFERENCES

[1] Z. Wu et al., "Scheduling-guided automatic processing of massive hyperspectral image classification on cloud computing architectures," *IEEE Trans. Cybern.*, vol. 51, no. 7, pp. 3588–3601, Jul. 2021.

[2] L. Zhang and L. Zhang, "Artificial intelligence for remote sensing data analysis: A review of challenges and opportunities," *IEEE Geosci. Remote Sens. Mag.*, vol. 10, no. 2, pp. 270–294, Jun. 2022.

[3] W. Liu, P. Duan, Z. Xie, X. Kang, and S. Li, "Learning from vision foundation models for cross-domain remote sensing image segmentation," *IEEE Trans. Image Process.*, vol. 34, pp. 4553–4565, 2025.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 05,2026 at 19:00:19 UTC from IEEE Xplore. Restrictions apply.

[4] L. Wu et al., "Modeling the label distributions for weakly-supervised semantic segmentation," *IEEE Trans. Pattern Anal. Mach. Intell.*, vol. 47, no. 8, pp. 6290–6306, Aug. 2025.

[5] Y. Liu, W. Guo, C. Yao, and L. Zhang, "Dual-perspective alignment learning for multimodal remote sensing object detection," *IEEE Trans. Geosci. Remote Sens.*, vol. 63, 2025, Art. no. 5404015.

[6] Y. Liu, X. Zhang, W. Guo, and L. Zhang, "Structured cross-resolution distillation for remote sensing salient object detection," *IEEE Trans. Geosci. Remote Sens.*, vol. 63, 2025, Art. no. 5637414.

[7] H. Chang, P. Wang, W. Diao, G. Xu, and X. Sun, "Remote sensing change detection with bitemporal and differential feature interactive perception," *IEEE Trans. Image Process.*, vol. 33, pp. 4543–4555, 2024.

[8] H. Zhang et al., "BiFA: Remote sensing image change detection with bitemporal feature alignment," *IEEE Trans. Geosci. Remote Sens.*, vol. 62, 2024, Art. no. 5614317.

[9] J. Lei, Y. Gu, W. Xie, Y. Li, and Q. Du, "Boundary extraction constrained Siamese network for remote sensing image change detection," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, 2022, Art. no. 5621613.

[10] X. Zheng, X. Chen, X. Lu, and B. Sun, "Unsupervised change detection by cross-resolution difference learning," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, pp. 1–16, 2022.

[11] J. Xu et al., "CGSL: Commonality graph structure learning for unsupervised multimodal change detection," *ISPRS J. Photogramm. Remote Sens.*, vol. 229, pp. 92–106, Nov. 2025.

[12] K. He, X. Zhang, S. Ren, and J. Sun, "Deep residual learning for image recognition," in *Proc. IEEE Conf. Comput. Vis. pattern Recognit.*, Jun. 2016, pp. 770–778.

[13] A. Dosovitskiy et al., "An image is worth 16x16 words: Transformers for image recognition at scale," 2020, *arXiv:2010.11929*.

[14] C. Han, C. Wu, M. Hu, J. Li, and H. Chen, "C2F-SemiCD: A coarse-to-fine semi-supervised change detection method based on consistency regularization in high-resolution remote sensing images," *IEEE Trans. Geosci. Remote Sens.*, vol. 62, 2024, Art. no. 4702621.

[15] Y. Wang, J. Sha, L. Gao, Y. Zhang, X. Rong, and C. Zhang, "A semi-supervised domain alignment transformer for hyperspectral images change detection," *IEEE Trans. Geosci. Remote Sens.*, vol. 61, 2023, Art. no. 5525411.

[16] Y. Tang, L. Zhang, W. Zhang, and Z. Jiang, "Meta-learning based semi-supervised change detection in remote sensing images," *IEEE Trans. Geosci. Remote Sens.*, vol. 63, pp. 1–15, 2025.

[17] L. Liu, D. Hong, L. Ni, and L. Gao, "Multilayer cascade screening strategy for semi-supervised change detection in hyperspectral images," *IEEE J. Sel. Topics Appl. Earth Observ. Remote Sens.*, vol. 15, pp. 1926–1940, 2022.

[18] K. Sohn et al., "FixMatch: Simplifying semi-supervised learning with consistency and confidence," in *Proc. Adv. Neural Inf. Process. Syst. (NIPS)*, 2020, pp. 596–608.

[19] X. Zhang, X. Huang, and J. Li, "Semisupervised change detection with feature-prediction alignment," *IEEE Trans. Geosci. Remote Sens.*, vol. 61, 2023, Art. no. 5401016.

[20] L. Ran, D. Wen, T. Zhuo, S. Zhang, X. Zhang, and Y. Zhang, "Adasemicd: An adaptive semi-supervised change detection method based on pseudo label evaluation," *IEEE Trans. Geosci. Remote Sens.*, vol. 63, 2025, Art. no. 5615814.

[21] J. Liu et al., "LGTC: Local–global tri-consistency network for semi-supervised change detection of remote sensing images," *IEEE Trans. Geosci. Remote Sens.*, vol. 63, 2025, Art. no. 5647213.

[22] A. Kirillov et al., "Segment anything," in *Proc. IEEE/CVF Int. Conf. Comput. Vis.*, Oct. 2023, pp. 4015–4026.

[23] L. Cai, H. Sun, X. Sun, H. Yan, and L. Gao, "HF-MCD: A heterogeneous fusion framework for multimodal change detection," *IEEE Trans. Geosci. Remote Sens.*, vol. 63, 2025, Art. no. 5407215.

[24] H. Zhang, K. Chen, C. Liu, H. Chen, Z. Zou, and Z. Shi, "CDMamba: Incorporating local clues into mamba for remote sensing image binary change detection," *IEEE Trans. Geosci. Remote Sens.*, vol. 63, 2025, Art. no. 4405016.

[25] Z. Lv, T. Yang, P. Zhong, W. Sun, J. Atli Benediktsson, and J. Li, "Hierarchical feature fusion triple network for change detection with bitemporal remote sensing images," *IEEE Trans. Geosci. Remote Sens.*, vol. 63, 2025, Art. no. 4401512.

[26] R. C. Daudt, B. Le Saux, and A. Boulch, "Fully convolutional Siamese networks for change detection," in *Proc. 25th IEEE Int. Conf. Image Process. (ICIP)*, Oct. 2018, pp. 4063–4067.

[27] H. Chen, Z. Qi, and Z. Shi, "Remote sensing image change detection with transformers," *IEEE Trans. Geosci. Remote Sens.*, vol. 60, pp. 1–14, 2022.

[28] S. Fang, K. Li, and Z. Li, "Changer: Feature interaction is what you need for change detection," *IEEE Trans. Geosci. Remote Sens.*, vol. 61, 2023, Art. no. 5610111.

[29] J. Pan, Y. Bai, Q. Shu, Z. Zhang, J. Hu, and M. Wang, "M-Swin: Transformer-based multiscale feature fusion change detection network within cropland for remote sensing images," *IEEE Trans. Geosci. Remote Sens.*, vol. 62, 2024, Art. no. 4702716.

[30] R. Hang, S. Xu, P. Yuan, and Q. Liu, "AANet: An ambiguity-aware network for remote-sensing image change detection," *IEEE Trans. Geosci. Remote Sens.*, vol. 62, 2024, Art. no. 5612911.

[31] M. Noman, M. Fiaz, H. Cholakkal, S. Khan, and F. S. Khan, "ELGC-Net: Efficient local–global context aggregation for remote sensing change detection," *IEEE Trans. Geosci. Remote Sens.*, vol. 62, 2024, Art. no. 4701611.

[32] D. Peng, L. Bruzzone, Y. Zhang, H. Guan, H. Ding, and X. Huang, "SemiCDNet: A semisupervised convolutional neural network for change detection in high resolution remote-sensing images," *IEEE Trans. Geosci. Remote Sens.*, vol. 59, no. 7, pp. 5891–5906, Jul. 2021.

[33] Y. Luo, B. Sun, S. Li, and Y. Hu, "Hierarchical augmentation and region-aware contrastive learning for semi-supervised semantic segmentation of remote sensing images," *IEEE Trans. Geosci. Remote Sens.*, vol. 63, 2025, Art. no. 4401311.

[34] L. Lv and L. Zhang, "ScaleMatch: Multi-scale consistency enhancement for semi-supervised semantic segmentation," in *Proc. AAAI Conf. Artif. Intell.*, 2025, vol. 39, no. 6, pp. 5910–5918.

[35] X. Zhang, X. Huang, and J. Li, "Joint self-training and rebalanced consistency learning for semi-supervised change detection," *IEEE Trans. Geosci. Remote Sens.*, vol. 61, 2023, Art. no. 5406613.

[36] Y. Zuo et al., "Robust instance-based semi-supervised learning change detection for remote sensing images," *IEEE Trans. Geosci. Remote Sens.*, vol. 62, 2024, Art. no. 4404815.

[37] K. Li et al., "SemiCD-VL: Visual-language model guidance makes better semi-supervised change detector," *IEEE Trans. Geosci. Remote Sens.*, vol. 63, 2025, Art. no. 5601613.

[38] N. Ravi et al., "SAM 2: Segment anything in images and videos," 2024, *arXiv:2408.00714*.

[39] Y. Qiao et al., "SAM enhanced semantic segmentation for remote sensing imagery without additional training," *IEEE Trans. Geosci. Remote Sens.*, vol. 63, 2025, Art. no. 5610816.

[40] X. Ma, Q. Wu, X. Zhao, X. Zhang, M.-O. Pun, and B. Huang, "SAM-assisted remote sensing imagery semantic segmentation with object and boundary constraints," *IEEE Trans. Geosci. Remote Sens.*, vol. 62, 2023, Art. no. 5636916.

[41] L. Ding, K. Zhu, D. Peng, H. Tang, K. Yang, and L. Bruzzone, "Adapting segment anything model for change detection in VHR remote sensing images," *IEEE Trans. Geosci. Remote Sens.*, vol. 62, 2024, Art. no. 5611711.

[42] X. Zhao et al., "Fast segment anything," 2023, *arXiv:2306.12156*.

[43] D. Zhang, F. Wang, L. Ning, Z. Zhao, J. Gao, and X. Li, "Integrating SAM with feature interaction for remote sensing change detection," *IEEE Trans. Geosci. Remote Sens.*, vol. 62, 2024, Art. no. 4513011.

[44] H. Chen and Z. Shi, "A spatial–temporal attention-based method and a new dataset for remote sensing image change detection," *Remote Sens.*, vol. 12, no. 10, p. 1662, May 2020.

[45] S. Ji, S. Wei, and M. Lu, "Fully convolutional networks for multisource building extraction from an open aerial and satellite imagery data set," *IEEE Trans. Geosci. Remote Sens.*, vol. 57, no. 1, pp. 574–586, Jan. 2019.

[46] M. Lebedev, Y. V. Vizilter, O. Vygolov, V. A. Knyaz, and A. Y. Rubis, "Change detection in remote sensing images using conditional adversarial networks," *Int. Arch. Photogramm.*, vol. 42, pp. 565–571, May 2018.

[47] E. D. Cubuk, B. Zoph, J. Shlens, and Q. V. Le, "Randaugment: Practical automated data augmentation with a reduced search space," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. Workshops (CVPRW)*, Jun. 2020, pp. 3008–3017.

[48] S. Mittal, M. Tatarchenko, and T. Brox, "Semi-supervised semantic segmentation with high- and low-level consistency," *IEEE Trans. Pattern Anal. Mach. Intell.*, vol. 43, no. 4, pp. 1369–1379, Apr. 2021.

[49] T.-H. Vu, H. Jain, M. Bucher, M. Cord, and P. Pérez, "Advent: Adversarial entropy minimization for domain adaptation in semantic segmentation," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit.*, Jun. 2019, pp. 2517–2526.

[50] X. Chen, Y. Yuan, G. Zeng, and J. Wang, "Semi-supervised semantic segmentation with cross pseudo supervision," in *Proc. IEEE/CVF Conf. Comput. Vis. Pattern Recognit. (CVPR)*, Jun. 2021, pp. 2613–2622.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 05,2026 at 19:00:19 UTC from IEEE Xplore. Restrictions apply.

[51] W. G. C. Bandara and V. M. Patel, "Revisiting consistency regularization for semi-supervised change detection in remote sensing images," 2022, *arXiv:2204.08454*.

[52] Y. Yang, X. Tang, J. Ma, X. Zhang, S. Pei, and L. Jiao, "ECPS: Cross pseudo supervision based on ensemble learning for semi-supervised remote sensing change detection," *IEEE Trans. Geosci. Remote Sens.*, vol. 62, 2024, Art. no. 5612317.



Liang Lv (Student Member, IEEE) received the Ph.D. degree in Earth exploration and information technology from China University of Geosciences, Wuhan, China, in 2025.

He is currently a Post-Doctoral Fellow with the School of Computer Science, Wuhan University. His main research interests include landslide detection, semantic segmentation, ensemble learning, and remote sensing.



Zilin Qin received the bachelor's degree from Nankai University, Tianjin, China, in 2022. He is currently pursuing the master's degree in computer science and technology with the School of Computer Science, Wuhan University, Wuhan, China.

His main research interests include computer vision, change detection, and remote sensing visual foundational models.



Xuefei Li received the Ph.D. degree from Wuhan University, Wuhan, China, in 2007.

He is currently a Professor with the School of Computer Science, Wuhan University. His research interests include reinforcement learning, satellite mission planning, and large-scale machine learning.



Lefei Zhang (Senior Member, IEEE) received the B.S. and Ph.D. degrees from Wuhan University, Wuhan, China, in 2008 and 2013, respectively.

He was a Big Data Institute Visitor with the Department of Statistical Science, University College London, London, U.K., and a Hong Kong Scholar with the Department of Computing, The Hong Kong Polytechnic University, Hong Kong, China. He is currently a Professor with the School of Computer Science, Wuhan University, and also with Hubei Luojia Laboratory, Wuhan. His research interests include pattern recognition, image processing, and remote sensing.

Dr. Zhang serves as an Associate Editor for IEEE TRANSACTIONS ON GEOSCIENCE AND REMOTE SENSING and an Associate Editor for IEEE GEOSCIENCE AND REMOTE SENSING LETTERS.

Authorized licensed use limited to: Hohai University Library. Downloaded on November 05,2026 at 19:00:19 UTC from IEEE Xplore. Restrictions apply.